

**SECRETARÍA DE EDUCACIÓN PÚBLICA
TECNOLÓGICO NACIONAL DE MÉXICO
INSTITUTO TECNOLÓGICO DE VERACRUZ**

ASIGNATURA:

Base de datos nosql

NOMBRE DE LA ACTIVIDAD:

Practica 1

ALUMNO:

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CARRERA:

INGENIERIA EN SISTEMAS COMPUTACIONALES

NÚMERO DE CONTROL:

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Reporte de Proyecto

Contenedor de Podman con MongoDB persistente

1. Objetivo

Crear un contenedor de Podman con una instancia de MongoDB, montando un volumen persistente, y demostrar que los datos almacenados sobreviven al apagar y encender el contenedor.

2. Herramientas utilizadas

- ❑ Podman Desktop (con máquina Podman sobre WSL2 en Windows)
- ❑ Imagen de MongoDB: `docker.io/mongodb/mongodb-community-server:latest`
- ❑ `mongosh` (shell de MongoDB, ejecutado dentro del contenedor)

3. Procedimiento y capturas requeridas

A continuación se detalla cada paso del procedimiento junto con la captura de pantalla que debe incluirse como evidencia.

Paso 1 — Creación del volumen persistente

En la imagen se ve que se creo y ya existe

```
PS C:\WINDOWS\system32> podman volume create mongo_datos
Error: volume with name mongo_datos already exists: volume already exists
PS C:\WINDOWS\system32> |
```

Paso 2 — Creación y arranque del contenedor

En la imagen se ve que se creo correctamente y ya existe

```
PS C:\WINDOWS\system32> podman run -d --name mongo-persistente -p 27017:27017 -v mongo_datos:/data/db:Z docker.io/mongo
db/mongodb-community-server:latest
Error: creating container storage: the container name "mongo-persistente" is already in use by 7ae122c6f81e662933b44c79b
77c0865a3c2201178d185e5ca2ffc636463d9c6. You have to remove that container to be able to reuse that name: that name is a
lready in use
PS C:\WINDOWS\system32> |
```

Paso 3 — Verificación de que el contenedor está corriendo

```
PS C:\WINDOWS\system32> podman ps
CONTAINER ID   IMAGE                                     COMMAND                  CREATED        STATUS        PORTS
7ae122c6f81e   docker.io/mongodb/mongodb-community-server:latest  mongod                  11 minutes ago Up 7 minutes  0.0.0.0:2701
->27017/tcp    mongo-persistente
PS C:\WINDOWS\system32>
```

Paso 4 — Conexión y creación de datos de prueba

```
7ae122c6f81e docker.io/mongodb/mongodb-community-server:latest mongod 11 minutes ago Up 7 minutes 0.0.0.0:27017
->27017/tcp mongo-persistente
PS C:\WINDOWS\system32> podman exec -it mongo-persistente mongosh
Current Mongosh Log ID: 6ab48e995c28313bdd8ce5af
Connecting to: mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.6.0
Using MongoDB: 8.3.11
Using Mongosh: 2.6.0
mongosh 2.12.0 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
The server generated these startup warnings when booting
2026-09-24T02:35:11.197+00:00: Using the XFS filesystem is strongly recommended with the WiredTiger storage engine. See http://dochub.mongodb.org/core/prodnotes-filesystem
2026-09-24T02:35:11.964+00:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
2026-09-24T02:35:11.964+00:00: For customers running the current memory allocator, we suggest changing the contents of the following sysfsFile
2026-09-24T02:35:11.964+00:00: For customers running the current memory allocator, we suggest changing the contents of the following sysfsFile
2026-09-24T02:35:11.964+00:00: We suggest setting the contents of sysfsFile to 0.
2026-09-24T02:35:11.964+00:00: We suggest setting swappiness to 0 or 1, as swapping can cause performance problems.
-----

test> |

test> use mi_bd
switched to db mi_bd
mi_bd> db.mi_tabla.insertOne({ nombre: "dato de prueba" })
{
  acknowledged: true,
  insertedId: ObjectId('6ab48c1641b36b1e328ce5b0')
}
mi_bd> db.mi_tabla.find()
[
  { _id: ObjectId('6ab48c1641b36b1e328ce5b0'), nombre: 'dato de prueba' }
]
```

Paso 5 — Apagado y encendido del contenedor

```
mi_bd> exit
PS C:\WINDOWS\system32> podman stop mongo-persistente
mongo-persistente
PS C:\WINDOWS\system32> podman start mongo-persistente
mongo-persistente
PS C:\WINDOWS\system32> podman exec -it mongo-persistente mongosh
Current Mongosh Log ID: 6ab48c6ce0904a27e38ce5af
Connecting to: mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.6.0
Using MongoDB: 8.3.11
Using Mongosh: 2.6.0
mongosh 2.12.0 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

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2026-09-24T02:35:11.964+00:00: We suggest setting the contents of sysfsFile to 0.
2026-09-24T02:35:11.964+00:00: We suggest setting swappiness to 0 or 1, as swapping can cause performance problems.
-----
```

Paso 6 — Verificación de persistencia

```
test> use mi_bd
switched to db mi_bd
mi_bd> db.mi_tabla.find()
[
  { _id: ObjectId('6ab48c1641b36b1e328ce5b0'), nombre: 'dato de prueba' }
]
mi_bd> exit
PS C:\WINDOWS\system32> podman volume create mongo_datos
Error: volume with name mongo_datos already exists: volume already exists
PS C:\WINDOWS\system32> podman run -d --name mongo-persistente -p 27017:27017 -v mongo_datos:/data/db:Z docker.io/mongo
db/mongodb-community-server:latest
Error: creating container storage: the container name "mongo-persistente" is already in use by 7ae122c6f81e662933b44c79b
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lready in use
PS C:\WINDOWS\system32> podman ps
CONTAINER ID   IMAGE                                     COMMAND                  CREATED        STATUS        PORTS
7ae122c6f81e   docker.io/mongodb/mongodb-community-server:latest   mongod                  11 minutes ago Up 7 minutes   0.0.0.0:27017
->27017/tcp    mongo-persistente
PS C:\WINDOWS\system32> podman exec -it mongo-persistente mongosh
Current Mongosh Log ID: 6ab48e995c28313bdd8ce5af
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2
.6.0
Using MongoDB:      4.3.11
```

4. Conclusión

Se demostró que, utilizando un volumen de Podman montado en /data/db, los datos de MongoDB persisten correctamente incluso después de detener y reiniciar el contenedor, cumpliendo con el objetivo del ejercicio.